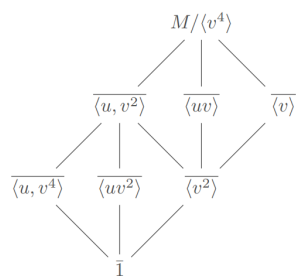




the ideas of ^{abstract} Algebra



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December 2025

THE IDEAS OF ALGEBRA

(in progress)

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CONTENTS

1. Introduction

2

1. INTRODUCTION

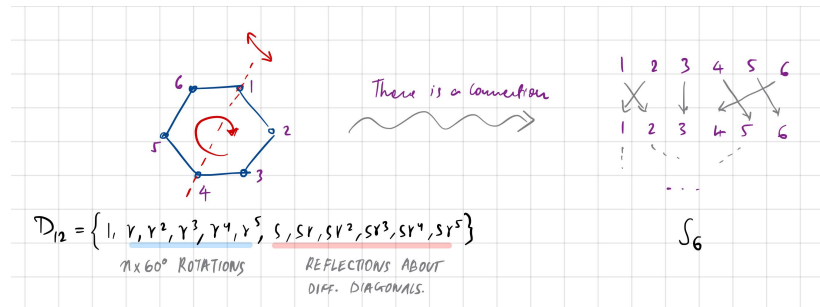
Abstract algebra is a subject in mathematics, that in some sense plays with some broken parts. It studies how these broken parts interact – with each other and perhaps with other objects too. The word algebra of course has a bit of history related exactly to this! See the below etymology from Google.



FIGURE 1. From Arabic al-jabr ‘the reunion of broken parts’, ‘bone-setting’, from jabara ‘reunite, restore’. The original sense, ‘the surgical treatment of fractures’, probably came via Spanish, in which it survives; the mathematical sense comes from the title of a book, ‘ilm al-jabr wa’l-muqābala ‘the science of restoring what is missing and equating like with like’, by the mathematician al-wārizmī

Our goal in this article is to appreciate the mechanics of a *group*, a type of algebraic object that captures precisely the mechanics of symmetries that we observe in nature or otherwise! A primary example one begins to probe these mechanics in, is a di-hedral group. Di-hedral groups (D_{2n}) are the symmetry groups of regular polygons. Through

reflections and rotations one can preserve the geometric structure of a regular polygon. But the *act* of preserving this information become an algebraic question, the broken pieces of reflections, rotations together grouping to become a – well, group. They have many interesting properties and we will relate them to another fundamental example called the Symmetric group (S_n) which are a collection of permutation one can make to any finite set or equivalently to the set $\{1, \dots, n\}$.



- Include Rigid motion exercises as a concrete demonstration of the theory. Isomorphisms and ACTION.